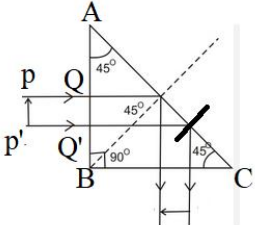
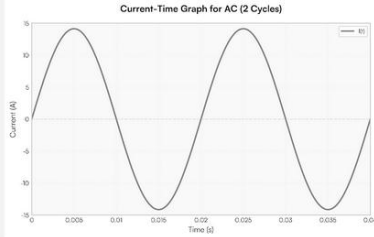


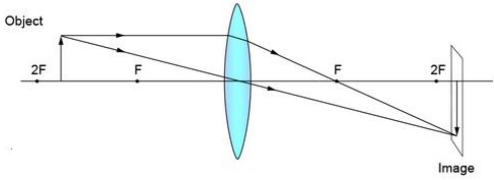
SEE 2082 (2026)
अनिवार्य विज्ञान तथा प्रविधि
उत्तरकुञ्जिका

यस कुञ्जिकामा भएको बाहेक अन्य सही वैकल्पिक तरिकाबाट समस्या समाधान गरेमा पनि अंक प्रदान गर्नुपर्ने छ ।
 उत्तरकुञ्जिकामा प्रत्येक चरणको प्राप्ताङ्क १ प्रदान गर्नुपर्नेछ तथा एकाइ छुट भएमा ०.५ अङ्क मात्र घटाउनुपर्ने छ ।

प्र. नं.	उत्तर	अंक
खण्ड 'क' (Section 'A')		
1 a.	(iii) Controlled variable	1
b	(ii) due to pressure produced in the ventricle of the heart is transmitted to the artery.	1
c	(iii) Caterpillar fungus	1
d	(iii) 8.34×10^{-18} N	1
e	(ii) Archimede's Principle	1
f	(ii) due to velocity of light changes when it enters into media having different densities.	1
g	(ii) Electromagnetic induction	1
h	(iv) Na_2SO_4	1
i	(iii) Reduction	1
j	(iv) Exhibits cleaning properties like soap but chemically different	1
खण्ड 'ख' (Section 'B')		
2.a	- Analog signal is continuous in nature while digital signal is in discrete in nature. Or any other correct point will get 1 mark	1
b	- Excessive consumption of fatty food. Or any other correct answer will get 1 mark	1
c	- Mountain B	1
d	- SI unit of mass is kilogram (kg) while SI unit of weight is Newton (N). Or any other correct point will get 1 mark	1
e	- Barometer - Or any other correct answer will get 1 mark.	1
f	- The transformer with more number of turns in the primary coil than in the secondary coil should be used. / step down.	1
g	- Faraday's Law of electromagnetic induction.	1
h	- Impure silver should be connected to the positive terminal (anode) and pure silver should be connected to the negative terminal (cathode).	1
i.	- Propene	1

	खण्ड 'ग' (Section 'C')	
3	Controlled variable- consumption of salt or any correct answer Dependent variable- blood pressure	1 1
4	- Similar- Both are vertebrates or any other correct points - Dissimilar- Fig A lay eggs, Fig-B gives directly birth or any other correct points	1 1
5	Class- Reptilia Comparison: Reptile has 3 chambered heart while Mammalia has 4 chambered heart.	
6	Drone $\xleftarrow{\text{Parthenogenesis}}$ 1 mark Bee $\xrightarrow{\text{Fertilization}}$ 1 Mark Worker bee	1 mark for each
7	- Mitosis cell division is called equational cell division. because the no. of chromosomes in daughter cell are same as the in the mother cells.	1 1
8	1. Parents will get child those who were facing problem in natural conception 2. It has higher success rate Or other correct answer gets similar marks.	1 1
9	- Due to global warming. - Due to excessive melting of ice Or other correct answer gets similar marks.	1 1
10	1. Holding of earth atmosphere 2. Establishment of building. Or other correct answer gets similar marks.	1 1
11		1 1
12		1 1
13	Expands forever, expansion slows but never stops Density: Less than critical density with suitable region. Or other correct answer gets similar marks.	1 1
14	$\text{MgO} + \text{HCl} \longrightarrow \text{MgCl}_2 + \text{H}_2\text{O}$ $\text{MgO} + 2\text{HCl} \longrightarrow \text{MgCl}_2 + \text{H}_2\text{O}$	1 1
15	IUPAC name: Ethane-1,2 di-ol It can be used as engine coolant. Or other correct answer gets similar marks.	1 1

16	- Calcium carbide (CaC₂) or Ethylene gas or Ethephone. - Its brings eye irritation or other correct answer gets similar marks	1 1						
खण्ड “घ” Section “D”								
17	<table> <tr> <th>Base band transmission</th> <th>Broad band transmission</th> </tr> <tr> <td> 1. Signal type: It has digital signal 2. frequency: It has low frequency range </td> <td> Signal type: It has analog signal frequency: It has high frequency range </td> </tr> </table> Two Suggestions: 1.Post authentic and quality content. 2. Respond politely to comments and feedback. Or, other correct answer gets similar marks	Base band transmission	Broad band transmission	1. Signal type: It has digital signal 2. frequency: It has low frequency range	Signal type: It has analog signal frequency: It has high frequency range	1 1		
Base band transmission	Broad band transmission							
1. Signal type: It has digital signal 2. frequency: It has low frequency range	Signal type: It has analog signal frequency: It has high frequency range							
18	a. When crossed between hybrid black rabbits (Bb) and pure white rabbits (bb) Chart b. Pure black (BB) and Pure white (bb) c. Definition: like, during the formation of gametes, the two alleles (factors) for a trait separate (segregate) from each other so that each gamete receives only one allele.	1 1 1 1						
19	a. Superior venacava b. It collects the deoxygenated blood c. <table> <tr> <th>C</th> <th>D</th> </tr> <tr> <td> 1. It represents pulmonary artery. 2. It carry deoxygenated blood </td> <td> 1. It represents pulmonery vein 2. It carry oxygenated blood </td> </tr> <tr> <td></td> <td>Or, other correct answer gets similar marks</td> </tr> </table>	C	D	1. It represents pulmonary artery. 2. It carry deoxygenated blood	1. It represents pulmonery vein 2. It carry oxygenated blood		Or, other correct answer gets similar marks	1 1 1 1
C	D							
1. It represents pulmonary artery. 2. It carry deoxygenated blood	1. It represents pulmonery vein 2. It carry oxygenated blood							
	Or, other correct answer gets similar marks							
20	a. In Himalayan region, volume of water increases due to increase in temperature according to anomalous expansion of water. b. Covering (insulating) pipes helps prevent freezing and bursting in winter. c. Due to anomalous expansion of water, water has maximum density at 4⁰C. So water having maximum density lie at the bottom of ponds. This temperature is suitable for all aquatic animals to survive.	1 1 2						

21	a. Ray diagram  b. i. Image is real ii. Image is inverted c. Image will be magnified	1 1 1 1
22	a. M is very active non-metal b. $1s^2, 2s^2, 2p^6, 3s^2 3p^6$ c. Y has larger atomic size. It has less nuclear force of attraction than X. So Y can lose faster than X. So Y is more reactive. d. Z= Mg, M= Cl $\text{Mg} + \text{Cl}_2 \longrightarrow \text{MgCl}_2$	1 1 1 1
23	a. Ammonia Gas (NH_3 gas) b. It helps to absorb the moisture and forms dry and pure ammonia gas. c. Ammonium sulphate is formed $2\text{NH}_3(\text{g}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow (\text{NH}_4)_2\text{SO}_4(\text{s})$	1 1 1 1